

The Direct-Sum Input from the Older MPS Representation Paper

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1 Source statement

The finite-length direct-sum input used behind the block-injective canonical-form route is not proved in the 2016 MPDO source. The 2016 paper cites the older representation paper for the block-injective-after-blocking step [CPGSV16, lines 340–345]. The local source for the older argument is `Papers/quant-ph_0608197/MPSarchive.tex`.

That source first introduces the injectivity condition *C1*: for some L_0 , the map

$$\Gamma_{L_0}(X) = \sum_{i_1, \dots, i_{L_0}} \text{tr}(X A_{i_1} \cdots A_{i_{L_0}}) |i_1, \dots, i_{L_0}\rangle$$

is injective. It immediately identifies this with the exact-length word products of length L_0 spanning the full matrix algebra [PGVWC07, lines 888–908].

The technical algebraic lemma then says that, for nonzero $D \times D$ matrices C , every matrix X is a finite sum

$$X = \sum_i R_i C S_i.$$

The proof reduces C to a matrix unit by elementary matrix operations and then reconstructs X from permuted copies of that matrix unit [PGVWC07, lines 1333–1344].

The next lemma is the direct-sum statement. Its local hypotheses are important: the tensor has $b \geq 2$ canonical blocks $A^1, \dots, A^b$, each block satisfies *C1*, the common length is $L_0 = \max_j L_0^{A^j}$, the block sizes are ordered, and the block states are assumed pairwise different [PGVWC07, lines 1321–1330]. Under these hypotheses, if $L \geq 3(b-1)(L_0+1)$, then

$$\sum_{j=1}^b \mathcal{G}_L^{A^j}$$

is a direct sum [PGVWC07, lines 1346–1408]. The proof is by induction on the number of blocks. The base case uses *C1* and the matrix-factorization lemma to

show that a nontrivial linear dependence would force inclusion, then equality, of the two length- L_0+1 MPS subspaces, contradicting uniqueness of the associated injective parent-space characterization. The induction step isolates one block with a local projector, applies the induction hypothesis to the remaining blocks, and then uses *C1* plus the factorization lemma to force the remaining coefficient to vanish.

2 MPS translation

For the non-periodic Fundamental Theorem route, the mathematical consequence is homogeneous and positive-length. After the canonical-form and basis-of-normal-tensors reduction, the selected representatives should admit one length at which their simultaneous word evaluations separate the direct-sum blocks. In trace-dual form, if

$$\sum_{j=1}^g \text{tr}(\Delta_j A_j^w) = 0 \quad \text{for every word } w \text{ of length } L,$$

then each Δ_j is zero. This is the finite-length direct-sum span input needed by the BNT pair-separation route.

This statement is not a consequence of the empty word and is not a cumulative span statement. The exact length L is part of the content. Therefore the formal proof should not replace the source input by a hypothesis that only says the cumulative pair algebra is full, nor by a theorem that uses the length-zero identity as padding.

3 Formal boundary

The matrix-factorization content of Lemma `lem1` from the source paper is now formalized in two steps.¹ The first says that, for any nonzero matrix C , the set of products $\{RCS : R, S \in M_D(\mathbb{C})\}$ spans $M_D(\mathbb{C})$. The second translates this into the relevant word-product form: if a tensor is L -block injective and $X \neq 0$, then $\text{span}\{A_u X A_v : |u| = |v| = L\} = M_D(\mathbb{C})$.

The next part of the two-block direct-sum argument is also formalized. The three-block trace relation forces, in the size-ordering convention $D_B \leq D_A$, an inclusion $\mathcal{G}_L^A \subseteq \mathcal{G}_L^B$ of length- L image spaces. Since block injectivity identifies $\dim \mathcal{G}_L^A = D_A^2$ and $\dim \mathcal{G}_L^B = D_B^2$, the inclusion and the size ordering together force $D_A = D_B$ and $\mathcal{G}_L^A = \mathcal{G}_L^B$.² The strict-size branch - if $D_B < D_A$, no nonzero coefficient on the larger block can satisfy the homogeneous three-block trace relation - is formalized separately,³ as is the conditional directness result: any

¹`Matrix.span`range`mul`nonzero`mul`eq`top` for the abstract matrix-algebra form; `MPSTensor.span`range`evalWord`mul`nonzero`mul`evalWord`eq`top` for the word-product translation.

²`MPSTensor.bondDim`eq`and`groundSpace`eq`of`three`block`trace`relation`left`of`dim`ge`.`

³`MPSTensor.not`three`block`trace`relation`left`of`isNBkInjective`of`dim`gt`;` `MPSTensor.pairTraceSeparatingAt`threeBlock`of`isNBkInjective`of`dim`gt`` for the trace-separation form;

external proof of the contradiction $\neg(D_A = D_B \wedge \mathcal{G}_L^A = \mathcal{G}_L^B)$ gives $\mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B = \{0\}$.⁴

The equal-size uniqueness branch is isolated as well. For injective tensors, equal length- L image spaces force equality of the periodic MPV lines. A supplied distinct-MPV-line hypothesis therefore rules out the equal-size collapse.⁵ The analytic source of distinct-MPV witnesses is formalized for irreducible trace-preserving blocks with normalized self-overlaps: if two blocks are not gauge-phase equivalent, then at arbitrarily large chain length their MPV states are not pointwise proportional.⁶ Combining the uniqueness argument with the finite-dimensional image-space step gives: if any chain length $N \geq \max\{2, L\}$ witnesses non-proportionality, then $\mathcal{G}_{3L}^A \cap \mathcal{G}_{3L}^B = \{0\}$.⁷ Zero intersection of image spaces at a fixed length, together with injectivity of the boundary-to-chain maps at that length, implies homogeneous pair trace separation.⁸ The two branches are combined for all ordered pairs of distinct blocks of a BNT family at the common length $3L$, and the result is fed into the selector and product-span construction.⁹

The already-injective BNT conclusion should now be stated either from `IsNormalCanonicalFormBNT` together with the required injective-block hypotheses, or directly from explicit separated-block hypotheses. In that form, when the selected BNT blocks are already injective at the required common length, the exact-length direct-sum span is a theorem rather than a remaining identity-padding assumption.

The downstream consumers of the direct-sum input are threefold. The exact-length span of a direct-sum block family and its two-block trace-dual form are expressed as predicates used throughout the pair route.¹⁰ Theorems in the period-window module turn a positive homogeneous window of the pair-span predicate into homogeneous trace separation.¹¹ BNT-level reformulations turn pairwise period-window data into a common exact-length span for a canonical-form BNT family.¹²

`MPSTensor.pairTraceSeparatingAt`threeBlock`of`isNBkInjective`of`dim`ne` for the convention-free version.

⁴`MPSTensor.groundSpace`inf`eq`bot`of`not`bondDim`eq`and`groundSpace`eq`of`dim`ge.`

⁵`MPSTensor.not`bondDim`eq`and`groundSpace`eq`of`mpvSubmodule`ne.`

⁶`MPSTensor.exists`ge`not`forall`mpv`eq`mul`of`not`gaugePhaseEquiv`of`irreducible`TP; MPSTensor.exists`ge`not`forall`mpv`eq`mul`of`blocksNotGaugePhaseEquiv`of`irreducible`TP` for the BNT-family form.

⁷`MPSTensor.groundSpace`inf`eq`bot`of`exists`not`forall`mpv`eq`mul`of`dim`ge; MPSTensor.groundSpace`inf`eq`bot`of`blocksNotGaugePhaseEquiv`sames`dim`of`dim`ge` for the BNT-separation packaging.

⁸`MPSTensor.pairTraceSeparatingAt`of`groundSpace`inf`eq`bot`of`isNBkInjective; and the combined forms MPSTensor.pairTraceSeparatingAt`threeBlock`of`exists`not`forall`mpv`eq`mul`of`dim`ge, MPSTensor.pairTraceSeparatingAt`threeBlock`of`blocksNotGaugePhaseEquiv`sames`dim`of`dim`ge.`

⁹`MPSTensor.forall`pairTraceSeparatingAt`threeBlock`of`blocksNotGaugePhaseEquiv; MPSTensor.exists`forall`pairTraceSeparatingAt`threeBlock`of`blocksNotGaugePhaseEquiv.`

¹⁰`WordTupleSpanTop, PairWordTupleSpanTop, PairTraceSeparatingAt.`

¹¹`TNLean/MPS/MPDO/BiCFDerivation/PairHomogenization.lean.`

¹²`TNLean/MPS/CanonicalForm/BlockDiagonalCommutant.lean.`

4 Conclusion

The formalized theorems cover the $L_0 = 1$ case: when the selected BNT blocks are already injective at length one (as supplied by the one-copy normal-canonical BNT formulation together with irreducible-block hypotheses), their pairwise not-gauge-phase-equivalence data and the fixed lengths from one-site injectivity imply the homogeneous pair-trace separation and product-word-span conclusions needed downstream. For this special case the formal result matches the paper’s conclusion: distinct BNT representatives yield a direct-sum span at a common finite length. However, it does not formalize the full generality of the paper’s direct-sum theorem. The older representation paper states the result for arbitrary L_0 , ordered block sizes, and $b \geq 2$ blocks; those features are replaced here by one-copy normal-canonical BNT and explicit separated-block hypotheses. The general- L_0 case, in which each normal block needs a Wielandt-type blocking step to reach length- L_0 injectivity, is not covered by these theorems alone; it requires the separate bounded and common injective-blocking results. The old warning against replacing exact-length span by cumulative span therefore remains mathematically important for that general setting.

For the full after-blocking normal-CF-BNT route, the conclusion is still conditional. The bounded and common injective-blocking results now supply the Wielandt part of the input, but the final CPSV theorem is not yet unconditional: the proof still has to transport the blocked BNT separation hypotheses through the chosen reblocking/reindexing maps and feed the blocked not-GPE facts into the BNT sector-data theorem `MPSTensor.ft`sector`bnt`equal`sector`data`. That transport is the remaining bridge between the already-injective direct-sum theorem surface and the canonical-form after-blocking comparison.

Accordingly, the blueprint should present the David/Pérez-García exact-length direct-sum input as formalized for the already-injective BNT specialization, while keeping the after-blocking CPSV statement visibly conditional until the reblocking transport and blocked BNT-separation handling have landed.

References

- [CPGSV16] J. Ignacio Cirac, David Pérez-García, Norbert Schuch, and Frank Verstraete. Matrix product density operators: Renormalization fixed points and boundary theories. Preprint; published version: *Annals of Physics* 378 (2017), 100–149, 2016.
- [PGVWC07] David Pérez-García, Frank Verstraete, Michael M. Wolf, and J. Ignacio Cirac. Matrix product state representations. *Quantum Information and Computation*, 7(5–6):401–430, 2007.